Absolute value inequalities

1

a $-5 \le x \le 5$

b

2 If $c > 0$, then $|u| < c$ is equivalent to $-c < u < c$.

3

a $-2 < x < 2$

b $x \ge 5, x \le -5$

4

a $|x| \le 7$

b $|x| > 9$

c $|2x + 3| \le 0.6$

5

a $|x| < 8$

b $|x| > 8$

c $|x - 8| \ge 2$

d $|x - (-2)| \le 5$

6

a

b

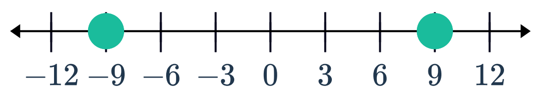
c

e

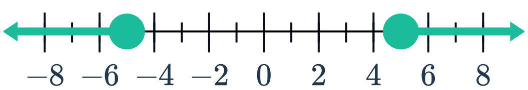
f No solution.

g

h



i



j No solution.

7 $(-\infty, \infty)$

8

a $x \geq 9, x \leq 5$

b $x > 16, x < -6$

c $-5 \leq x \leq 13$

d $-8 < x < 4$

e $-10 \leq x \leq 4$

f $-\dfrac{16}{3} < x < 2$

g $-4 \leq x \leq 1$

h $x \geq 7, x \leq -7$

i $x > 15, x < -15$

j $x < 3, x > 8$

k $x < -2, x > -\dfrac{2}{3}$

l $x \geq 15, x \leq -1$

m $x \leq -3$

Applications

9 $18.86 \leq n \leq 18.94$

10

a $h \geq 58.225, h \leq 41.775$

b If a coin is tossed 100 times and the number of outcomes that result in heads is 41 or less or 59 or more, then the coin is unfair.

11 $17.75 \leq m \leq 18.09$